

DSN X BCT LLM AGENT CHALLENGE

TASK A: USER MODELING

Persona-Driven User Modeling for Synthetic Review Generation Using Large
Language Model.

PROJECT HITORI

by

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the DSN X BCT LLM Agent Challenge

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Chapter 1

Abstract

User modeling is a fundamental task in recommender systems, enabling the prediction of user preferences and behaviours. Recent advances in large language models (LLMs) have enabled new paradigms for behavioural simulation beyond traditional collaborative filtering approaches. This work proposes a persona-driven user modeling framework that generates synthetic product reviews conditioned on personality traits, historical behaviour, and contextual product information.

The system integrates structured user representations with in-context learning to simulate both rating behaviour and natural language reviews. Experimental results show that while the model achieves moderate review quality, behavioural fidelity remains a key challenge.

Keywords: User Modeling, LLMs, Behaviour Simulation, Prompt Engineering

Chapter 2

Introduction

User modeling is a core component of modern recommender systems, enabling systems to predict user preferences and simulate decision-making behaviour. Traditional approaches rely heavily on collaborative filtering and matrix factorization techniques, which often struggle with cold-start problems and lack interpretability.

Recent advances in large language models have introduced new opportunities for simulating human-like behaviour using prompt-based learning rather than explicit training. These models can generate coherent text conditioned on structured inputs such as personality traits and contextual history.

2.1 Problem Statement

Despite progress in recommender systems, current models face key limitations:

- Inability to model nuanced personality-driven behaviour
- Poor generalization in cold-start scenarios

- Lack of interpretability in recommendation decisions

Research Objectives:

- To design a user modeling framework that captures personality-driven behaviour
- To simulate realistic review generation using LLMs
- To evaluate behavioural fidelity and rating consistency

Research Questions:

- Can LLMs simulate consistent user behaviour from structured personality traits?
- How does historical review context influence generated outputs?
- What factors improve behavioural fidelity in synthetic user modeling?

Chapter 3

Literature review

Traditional recommender systems rely on collaborative filtering and content-based approaches. While effective, they suffer from sparsity and cold-start issues.

Recent research has explored neural approaches and transformer-based architectures for recommendation tasks. However, most systems focus on prediction accuracy rather than behavioural realism.

Large language models have recently been applied to recommendation tasks through prompt engineering and in-context learning. These approaches allow models to simulate user preferences without explicit training.

Despite these advances, few works explicitly model personality-driven behaviour in review generation tasks. This work addresses that gap by integrating personality traits, behavioural filtering, and LLM-based generation.

Chapter 4

Methodology

4.1 Overview

The system simulates user behaviour using the following pipeline:

User Persona → Behaviour Filtering → Prompt Engineering → LLM Generation
→ JSON Parsing

4.2 System Architecture

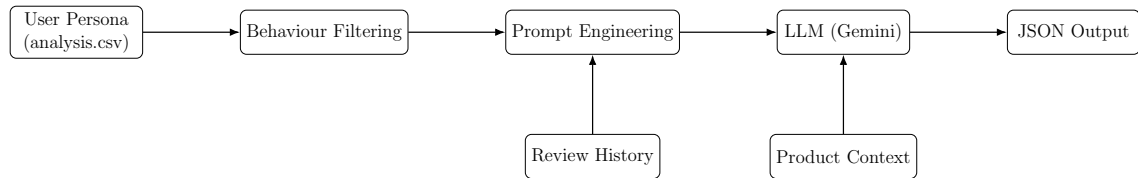


Figure 4.1: Architecture of the persona-driven user modeling system.

4.3 Key Components

4.3.1 User Modeling

User profiles are constructed using Big Five personality traits to simulate behavioural tendencies.

4.3.2 Behaviour Filtering

Real Amazon reviews are filtered based on sentiment and writing style alignment.

4.3.3 Prompt Engineering

Prompts combine personality traits, review history, and product context using in-context learning.

4.3.4 LLM Generation

The Gemini model generates ratings and reviews conditioned on structured prompts.

4.3.5 Output Parsing

Outputs are structured into JSON format for evaluation.

Chapter 5

Evaluation

5.1 Metrics

- Rating Accuracy
- Review Quality
- Behavioural Fidelity

5.2 Results Summary

- Average Rating: 4.2 (positive bias observed)
- Review Quality: Moderate realism
- Behavioural Fidelity: Weak alignment with persona

5.3 Discussion

While the system successfully generates coherent and plausible reviews, behavioural alignment remains limited. This suggests that LLMs require stronger conditioning mechanisms to fully capture personality-driven variation.

5.4 Future Work

Future improvements include:

- Fine-tuning personality-to-behaviour mapping
- Adding reinforcement learning from human feedback
- Multi-turn user simulation
- Embedding-based behavioural filtering